

Voice-Based Emotion Recognition in MATLAB Using MFCC Feature Extraction and Support Vector Machine

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Abstract— This paper presents a speech emotion recognition system designed to classify emotional states from speech signals using Mel-Frequency Cepstral Coefficients (MFCC) and a Support Vector Machine (SVM) with a radial basis function (RBF) kernel. The dataset consists of 535 audio samples representing seven emotion classes: anger, boredom, disgust, fear, happiness, and sadness. Prior to classification, feature normalization was applied, and the Synthetic Minority Over-sampling Technique (SMOTE) was used to address class imbalance in the training data. Model parameters were optimized using grid search with five-fold cross-validation. Experimental results show that the proposed system achieved a testing accuracy of 94.92%, with precision, recall, and F1-score values of 0.9558, 0.9488, and 0.9498, respectively. The confusion matrix analysis indicates that most emotion classes were classified correctly, with only minor misclassifications between acoustically similar emotions. These results demonstrate that the proposed MFCC–SVM approach provides reliable performance for speech emotion recognition.

Keywords— Speech Emotion Recognition, MFCC, Support Vector Machine, Voice Classification

I. INTRODUCTION

Advances in voice processing technology and artificial intelligence have encouraged more in-depth research into the ability of systems to recognize human emotional characteristics through vocal signals [1]. Emotions play an important role in communication because they influence how people convey information, make decisions, and interact with their environment [2]. However, automatic emotion identification remains a challenge due to the subjective and dynamic nature of emotions [1], [3], which are influenced by various linguistic and non-linguistic factors. Therefore, the use of voice recognition as an approach to distinguish human emotions offers significant opportunities in the fields of human-computer interaction, healthcare, education, and security systems.

Previous research shows that emotions in the voice can be identified through acoustic features such as pitch, energy, formants, and spectral characteristics that appear during emotional changes [3], [4]. Various machine learning and deep learning techniques have been applied to perform emotion classification, but system performance is still greatly

influenced by feature extraction quality, dataset characteristics, and the model's ability to handle speaker and environmental variations. These challenges highlight the need for a more effective and structured approach to processing voice signals for emotion recognition purposes.

In this study, voice recognition techniques are used to distinguish human emotions based on acoustic patterns and vocal characteristics that represent each emotion category. Signal processing, feature extraction, and model analysis are performed using MATLAB, which provides a stable numerical computing environment and various built-in functions that support signal processing and machine learning. The main objective of this research is to evaluate the system's ability to recognize emotions through voice and analyze the factors that affect its accuracy. The contributions of this research include a deeper understanding of the mapping of voice features to emotional states and the basis for developing a more adaptive and efficient emotion recognition system.

II. LITERATURE REVIEW

Voice recognition is a field that focuses on identifying vocal patterns to recognize specific characteristics of speech, whether in the form of words, speakers, or emotional states. Early research in emotion recognition made extensive use of basic acoustic features such as pitch, intensity, and speech tempo as indicators of emotional change [3]. These studies found that emotions such as anger, sadness, or joy exhibit significantly different spectral and prosodic patterns.

Subsequent research introduced more complex approaches using Mel-Frequency Cepstral Coefficients (MFCC), Linear Predictive Coding (LPC), and temporal energy features [4], [5]. These features proved to be effective because they were able to capture important characteristics of human voice production. Classification models such as Support Vector Machine (SVM), k-Nearest Neighbor (k-NN), and Gaussian Mixture Models (GMM) have been widely used for emotion recognition and have shown good performance on controlled datasets [6].

As technology advances, recent research has widely applied deep learning, particularly Convolutional Neural

Networks (CNN) and Recurrent Neural Networks (RNN), to automatically extract emotion patterns through spectrogram representation. However, this method typically requires large datasets, high computational resources, and complex training processes [1]. On the other hand, approaches based on classical feature extraction and machine learning models remain relevant because they are more concise, stable, and easy to implement in platforms such as MATLAB.

III. METHODOLOGY

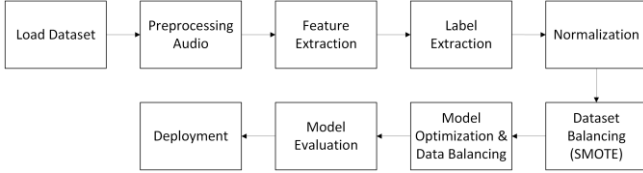


Figure 1 Block Diagram

The research methodology consists of five main stages: data collection, signal pre-processing, feature extraction, model training, and performance evaluation, as illustrated in the block diagram shown in Fig. 1. All stages were implemented using MATLAB, which provides comprehensive tools for signal processing and machine learning analysis.

A. Data Collection

The dataset used consists of voice recordings that have been labeled with emotions such as angry, sad, happy, afraid, and neutral. All voice files are in WAV format and are of good recording quality. The dataset is then divided into two parts, namely training data and the test data.

B. Signal Preprocessing

The preprocessing stage was carried out to ensure that the signals were in optimal condition before analysis. The steps taken included amplitude normalization to standardize sound intensity, noise reduction using filter techniques, and framing and windowing with Hamming windows. This stage was important so that the extracted features were more accurate and not affected by external interference [5].

C. Feature Extraction

Feature extraction is performed using several key acoustic features. Mel-Frequency Cepstral Coefficients (MFCC) features are used as the main features because they are capable of representing the spectral characteristics of human speech [4]. In addition, pitch, energy, and formant features are also extracted to capture signal variations related to emotional changes. MATLAB provides built-in functions such as mfcc, pitch, and lpc to support this process.

The main steps in MFCC extraction are described narratively as follows:

1. Framing

The continuous audio signal is divided into several short frames (25 ms) with an overlap between frames of 10 ms. The purpose is so that the signal can be considered stationary in each frame. The number of frames (Nf) can be calculated with:

$$L = T_f \times f_s$$

$$H = T_f \times f_s$$

$$x_k[n] = x[n + kH], 0 \leq n < L$$

Figure 2.2 Framing Formula

Where:

$x[n]$ = speech signal

f_s = sampling frequency

T_f = frame duration (e.g., 25 ms)

H = hop duration (e.g., 10 ms)

k = frame index

2. Filtering (Pre-emphasis)

This step aims to amplify the high-frequency components. A first-order linear filter is applied with:

$$y[n] = x[n] - \alpha x[n - 1]$$

Figure 2.3 Filtering Formula (Pre-emphasis)

With:

$\alpha \in [0.9, 0.97]$ (commonly 0.97)

$x[n]$ = input signal

$y[n]$ = filtered signal

This step improves the signal-to-noise ratio and maintains signal stability during Fourier transformation [5].

3. Windowing

Each frame is multiplied by the Hamming window function to reduce spectral leakage [6].

$$x_w[n] = x_k[n] \times w[n]$$

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{L-1}\right)$$

Figure 2.4 Windowing Formula

Where:

$x_w[n]$ = windowed frame

$w[n]$ = hamming window

$0 \leq n < L$

4. Fast Fourier Transform (FFT)

After windowing, the signal is converted to the frequency domain with FFT to obtain the magnitude spectrum [7]:

$$X[k] = \sum_{n=0}^{N-1} x_w[n] e^{-\frac{j2\pi kn}{N}}$$

$$P[k] = \frac{1}{N} |X[k]|^2$$

Figure 2.5 Fast Fourier Transform Formula

Where:

$X[k]$ = frequency domain

$P[k]$ = power spectrum

N = FFT size

k = frequency bin index

5. Mel-Scaling

The Mel scale is used to mimic human perception of frequency. The relationship between linear frequency (Hz) and the Mel scale :

$$Mel(f) = 2595 \log_{10} \left(1 + \frac{f}{700} \right)$$

$$f = 700 \left(10^{\frac{Mel}{2595}} - 1 \right)$$

Figure 2.6 Mel-Scaling Formula

Where:

Mel = mel scale

f = frequency in Hz

6. Mel Filter Bank

The FFT result is multiplied by triangular filter banks based on the Mel scale to calculate the energy of each frequency band :

$$E_m = \sum_{k=0}^{N-1} P[k] \times H_m[k]$$

$$H_m[k] = \begin{cases} 0 & k < f_{m-1} \\ \frac{k-f_{m-1}}{f_m-f_{m-1}} & f_{m-1} \leq k < f_m \\ \frac{f_{m+1}-k}{f_{m+1}-f_m} & f_m \leq k < f_{m+1} \\ 0 & k \geq f_{m+1} \end{cases}$$

$$\hat{E}_m = \log(E_m)$$

Figure 2.7 Mel Filter Bank Formula

Where:

E_m = mel energy

$H_m[k]$ is the m th filter response.

m = filter index

f_m = center frequencies on Mel scale

= log mel energy

7. Discrete Cosine Transform (DCT)

The energy from the Mel filter banks is transformed using DCT to produce cepstral coefficients:

$$MFCC_n = \sum_{m=1}^M \hat{E}_m \cos \left[\frac{\pi n}{M} (m - 0.5) \right]$$

Figure 2.8 Discrete Cosine Transform Formula

Where:

M = number of Mel filters

n = MFCC coefficient index

The final result cn is the MFCC Output used as a feature for classification .

D. Class Labeling

Emotion labels were assigned automatically based on predefined codes embedded in the audio file names. Each label corresponds to one of seven emotion classes: anger, boredom, disgust, fear, happiness, sadness, and neutral.

E. Data Balancing Using SMOTE

Due to the imbalance in the emotion class distribution, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to the extracted feature set. SMOTE generates synthetic samples for minority classes based on k -nearest neighbors ($k = 5$), resulting in a balanced dataset and reducing classification bias.

F. Classification and Parameter Optimization

The classification model was trained using a Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel in a one-vs-all multi-class configuration. Given a feature vector $x \in \mathbb{R}^d$, the RBF kernel function used in this study is defined as:

$$K(x, x') = \exp(-\gamma \|x - x'\|^2)$$

Where,

exp: Exponential curve function

γ : Controls how far influence of a point spreads

$\|x - x'\|^2$: Squared distance between points

Hyperparameter optimization was performed using grid search combined with 5-fold cross-validation on the balanced dataset obtained through SMOTE. The parameter γ , corresponding to the KernelScale parameter in the

implementation, and the regularization parameter C (BoxConstraint) were evaluated within the range of 1 to 10. The parameter combination that achieved the highest cross-validation accuracy was selected as the optimal configuration.

Subsequently, the dataset was divided into 80% training data and 20% testing data using a hold-out validation scheme. The final SVM model was then trained on the training set using the selected hyperparameters.

G. Model Training and Evaluation

The performance of the trained model was evaluated using the testing dataset to assess its generalization capability [6]. Several evaluation metrics were employed, including accuracy, confusion matrix, precision, recall, and F1-score. Accuracy was used to measure the overall classification correctness, while the confusion matrix provided detailed insight into class-wise prediction results and misclassification patterns. Precision and recall were utilized to evaluate the reliability and completeness of predictions for each emotion class, respectively. The F1-score, defined as the harmonic mean of precision and recall, was used to provide a balanced performance measure, particularly in the presence of class imbalance. Together, these metrics offer a comprehensive evaluation of the proposed speech emotion recognition system.

IV. TESTING RESULTS

This section presents the experimental results of the proposed speech emotion recognition system, including audio signal analysis across emotion classes, model optimization, and classification performance evaluation.

A. Model Training and Evaluation

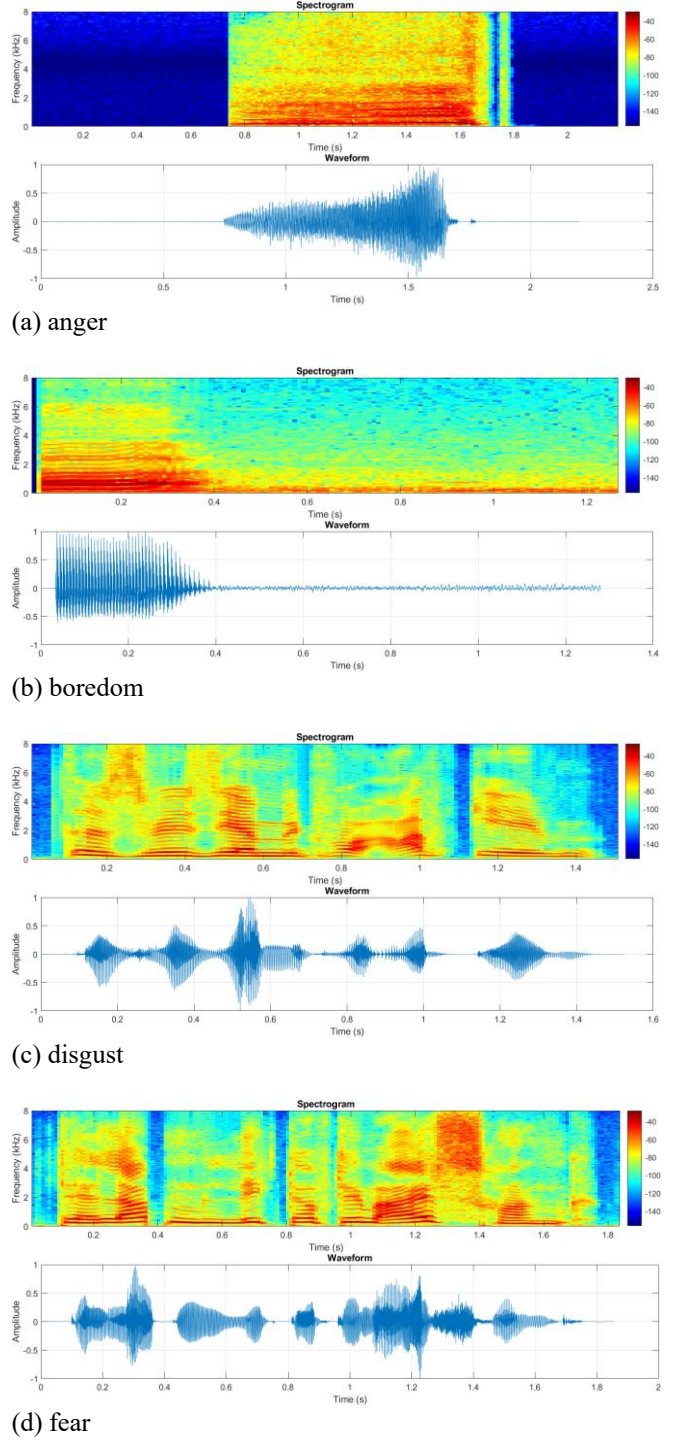
The experiments were conducted using a dataset consisting of 535 audio files, representing seven emotion classes: anger, boredom, disgust, fear, happiness, neutral, and sadness. Feature extraction was performed using Mel-Frequency Cepstral Coefficients (MFCC) to capture the perceptually relevant spectral characteristics of speech signals. Feature normalization was applied prior to classification to ensure uniform feature scaling.

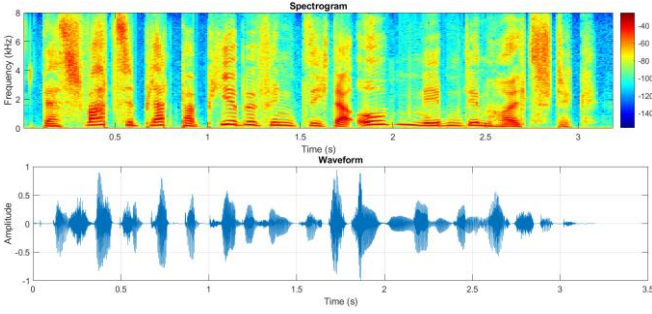
To address class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training data. The dataset was divided using a hold-out validation strategy, where 80% of the data was used for training and 20% for testing.

B. Waveform and Spectrogram Analysis of Emotional Speech

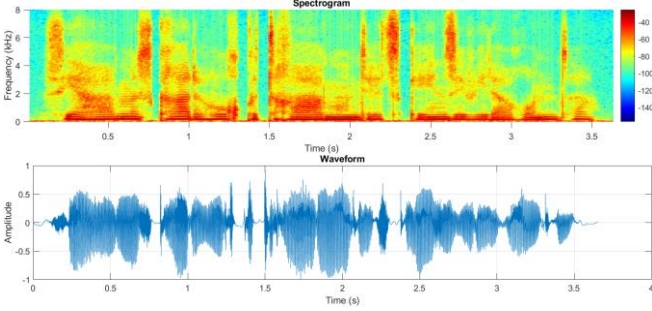
Fig. 2 presents the time-domain waveforms and corresponding spectrograms of speech signals for all emotion classes used in this study. These visualizations highlight the acoustic variations across different emotional states.

Fig. 2. Time-domain waveforms and spectrograms of speech signals for all emotion classes: (a) anger, (b) boredom, (c) disgust, (d) fear, (e) happiness, (f) sadness.





(e) happiness



(f) Sadness

Each emotion class exhibits distinctive acoustic characteristics. Emotions such as anger and fear demonstrate higher amplitude fluctuations and broader spectral energy distributions, indicating increased vocal intensity. In contrast, boredom and neutral emotions show relatively stable waveforms with lower energy concentration. Emotions such as disgust and sadness exhibit moderate amplitude variations with more localized spectral patterns. These observable differences support the effectiveness of MFCC features in capturing discriminative emotional characteristics from speech signals.

C. SVM Parameter Optimization

The classification model employed in this study is a Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel. Hyperparameter optimization was conducted using Grid Search with five-fold cross-validation to determine the optimal values of KernelScale and BoxConstraint.

The optimal parameters obtained from the grid search process are summarized in Table 1.

Table 1 Optimal SVM Parameters Obtained Using Grid Search

Parameter	Value
KernelScale	9
BoxConstraint	9
CV accuracy (%)	94.15

The cross-validation accuracy of 94.15% indicates that the optimized SVM model demonstrates strong generalization capability during training.

D. Classification Performance Evaluation

The optimized SVM model was evaluated using the testing dataset, achieving an overall testing accuracy of 94.92%. Additional evaluation metrics including precision, recall, F1-score, and loss were computed to provide a comprehensive performance assessment. The results are summarized in Table 2.

Table 2 Testing Performance Metrics

Metric	Value
Accuracy	94.92%
Precision	0.9558
Recall	0.9488
F1-score	0.9498
Loss	0.0508

The results indicate consistent and balanced classification performance across all emotion classes.

E. Confusion Matrix Analysis

The confusion matrix of the testing results is shown in Fig. 3, illustrating the classification performance for each emotion class.

Fig. 3. Confusion matrix of the proposed MFCC–SVM-based speech emotion recognition system.

Confusion Matrix (Akurasi: 94.92%)								
True Class	anger	25			1			
	boredom		24					1
	disgust			23	2			
	fear				26			
	happiness	3			1	21		
	neutral	1					24	
	sadness							25
		anger	boredom	disgust	fear	happiness	neutral	sadness
		Predicted Class						

The confusion matrix demonstrates that most samples are correctly classified, as reflected by the dominant diagonal elements. Minor misclassifications are observed, particularly between emotions with similar acoustic characteristics, such as happiness and anger, or disgust and fear. However, these errors occur infrequently and do not significantly affect the overall system performance.

V. CONCLUSION

This study presented a speech emotion recognition system based on Mel-Frequency Cepstral Coefficients (MFCC) and a Support Vector Machine (SVM) with an RBF kernel implemented in MATLAB. Using a dataset of 535 speech samples across seven emotion classes, the proposed system successfully extracted discriminative acoustic features and classified emotional states with reliable performance. The application of feature normalization and SMOTE effectively addressed data imbalance, while grid search with five-fold cross-validation ensured optimal model parameter selection.

Experimental evaluation demonstrated that the proposed system achieved a testing accuracy of 94.92%, with precision, recall, and F1-score values of 0.9558, 0.9488, and 0.9498, respectively. The confusion matrix analysis showed that most emotion classes were correctly classified, with minor misclassifications occurring primarily between emotions with similar acoustic characteristics. These results confirm that the MFCC–SVM approach provides a robust and effective baseline for speech emotion recognition. Future work may focus on improving generalization by incorporating larger and more diverse datasets, additional prosodic features, or more advanced classification techniques.

VI. REFERENCES

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